

Hang a Cabinet Door on Hinges SOP (1x Episode: 2 Hinges)

One episode hangs one door on both hinges. The table begins with the **frame board** lying flat across the back of the table, just right of the center, top face up. The **door** lies flat in front of it on the **door spot**, back face up, hinge edge toward the board. Two **cup holes** are already bored in the door's back face. Two **plate stations** are already marked on the board's top face. Every screw hole is a pre-bored pilot hole.

The order never changes: screw one hinge into each cup hole, clip a plate onto each hinge arm, slide the door back to offer it to the board, screw both plates down, adjust the cams, then swing-test the door. Nothing is screwed to the board until both plates are clipped home. No cam is turned until both plate screws are seated. The episode does not end until the door has swung open and closed twice with both hinges holding.

The right gripper does all the tool work: it holds the electric screwdriver for the four mounting screws, then the hand driver for the cams, and it makes the swing test. The left gripper does the supporting work: it presents the hinges, the plates, and the screws, holds the door down while screws are driven, slides the door into place, and holds the frame board flat during the swing test. The left gripper stays on the left of the door and never reaches across the board.

The table is set up in one of three ways. Only the **parts tray** moves; the frame board, the door, the screw holder, and the two tool holders are in the same place in all three.

- **Config L1:** the parts tray is at the front-left, beside the screw holder.
- **Config L2:** the parts tray is at the back-left, left of the frame board's left end.
- **Config M:** the parts tray is at the front-center, in front of the door's free edge.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the parts tray starts at the front-left (**Config L1**), the back-left (**Config L2**), or the front-center (**Config M**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the **left gripper** takes every hinge and every plate from the tray in all three configs; the tray is always on its side of the table. No arm reaches across the table.
- **Fixed roles:** the right gripper holds the electric screwdriver from Step 1 through Step 6, while every part is picked, so it never fetches from the tray. That is why no right-side zone is used, and why the tool zone, the screw holder, the door spot, and the frame board do not move. The order of the steps is the same in all three configs.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the frame board with both plate stations, the door on the door spot, the parts tray in its start zone, the screw holder at the front-left, and the two tool holders at the right.
3. The frame board and the door are both visible from above, so both cup holes, both plate stations, the hinge gap, the line marks, and both cams on each hinge can be seen.
4. The electric screwdriver has the correct bit, is set to forward and low speed, and its clutch is at the validated setting for these screws.
5. Both arms are at home with grippers open.
6. The table is bare apart from the frame board, the door, the parts tray, the screw holder, the two tools, and robot hardware.
7. The right arm reaches both cup holes, both plate stations, both hinges' cams, the door's front edge, and both tool holders without stretching. The left arm reaches the parts tray in its start zone — front-left (Config L1), back-left (Config L2), or front-center (Config M) — the screw holder, the door's left half, and the left end of the frame board without stretching.

Materials checklist

1. One **frame board** lies flat across the back of the table, just right of the center, top face up and square to the front edge.
2. Two **plate stations** are marked on the board's top face, just behind its front edge: **station 1** on the left and **station 2** on the right. Each station has one pre-bored pilot hole.
3. Two **line marks** are on the board's front edge, one under where the door's left end belongs and one under where its right end belongs.
4. One **door** lies flat on the **door spot**, in front of the board and right of the center, **back face up**, with its **hinge edge** toward the board and its **free edge** toward the front edge.
5. Two **cup holes** are bored in the door's back face near the hinge edge: **cup hole 1** on the left and **cup hole 2** on the right. Each has a **pin hole** and one pre-bored pilot hole beside it.
6. The **parts tray** holds two **hinges**, cups up and arms flat, and two **plates**. It stands in one of three places:
 - **Config L1:** front-left, beside the screw holder

- **Config L2:** back-left, left of the frame board's left end and clear of it
- **Config M:** front-center, in front of the door's free edge, clear of the door and the tool zone

- The **screw holder** at the front-left holds four matching screws, head-up.
- The **electric screwdriver** sits in its **cradle** at the front-right. The **hand driver** sits in its **holder** beside the cradle.
- All parts are clean and undamaged: no stripped pilot holes, no burred cup rims, no bent hinge arms, no chewed screw heads, and both cams turn by hand.
- Keep the rest of the left side of the table clear. The left gripper works from there onto the tray, the holder, and the door's left half.
- Before collection, confirm by hand that each cup drops into its cup hole with the pin in the pin hole, that a plate clips onto each arm with a click, that sliding the door back sets both plates flat on the board with their screw holes over the pilot holes, that both cams move the door when turned, and that the door swings up to upright and back flat without binding.

Workspace layout

- **Back of the table, just right of center:** the frame board — it stays here all episode and is never lifted
- **Front edge of the board:** the two line marks; just behind it, plate station 1 on the left and plate station 2 on the right
- **Door spot:** in front of the board, right of center — the door starts here, back face up
- **Start zone:** the parts tray with two hinges and two plates — front-left beside the screw holder (Config L1), back-left left of the frame board (Config L2), or front-center in front of the door (Config M)
- **Front-left:** the screw holder with four screws
- **Front-right tool zone:** the screwdriver cradle and the hand driver holder
- **Left side:** kept clear apart from the tray in Config L2, so the left gripper can come in onto the tray, the holder, and the door

Arm assignments

- **Left gripper:** picks the hinges and plates from the parts tray — front-left in Config L1, back-left in Config L2, front-center in Config M — and the screws from the screw holder, and presents them; presses the door down while cup screws are driven; presses each hinge arm down while its plate screw is driven; slides the door back to offer it to the board; and holds the frame board flat during the swing test, so nothing moves while the right gripper works.
- **Right gripper:** holds the electric screwdriver and drives all four mounting screws, holds the hand driver and turns all four cams, and lifts and lowers the door for the swing test. It does the same in all three configs and never fetches from the tray.

Vocabulary

- **Start zone:** where the parts tray stands at the start of the episode — front-left (**Config L1**), back-left (**Config L2**), or front-center (**Config M**). One per episode, chosen before recording and never changed mid-episode.
- **Frame board:** the flat board the door is hung on. It stays at the back of the table all episode and is never lifted.
- **Door:** the flat panel being hung. Its **back face** is the face the hinges screw onto and is up all episode. Its **show face** is against the table.
- **Hinge edge:** the edge of the door nearest the frame board, where the cup holes are.
- **Free edge:** the edge of the door nearest the front edge of the table. This is the edge the right gripper lifts for the swing test.
- **Cup hole:** the round hole bored in the door's back face that a hinge cup drops into.
- **Pin hole:** the small hole beside a cup hole that the hinge's locating pin drops into. With the pin in, the hinge cannot turn in the cup hole.
- **Hinge:** one piece made of a cup and an arm.
- **Cup:** the round part of the hinge that drops into a cup hole.
- **Locating pin:** the small peg sticking out of the cup that drops into the pin hole.
- **Arm:** the flat strap running from the cup toward the frame board. It carries both cams and takes the plate at its far end.
- **Plate:** the small block that screws to the frame board. It clips under the far end of an arm.
- **Plate station:** a marked spot on the board's top face with one pilot hole, where a plate is screwed down. Station 1 is the left one, station 2 the right one.
- **Clicked home:** the plate has been pressed onto the arm until it clicks and then does not rock, lift, or slide off the arm when the door is moved.
- **Hinge gap:** the open strip between the door's hinge edge and the board's front edge.
- **Line marks:** the two marks on the board's front edge that the door's left and right ends line up with.
- **Line cam:** the cam nearer the cup. Turning it slides the door left or right along the board.
- **Gap cam:** the cam nearer the plate. Turning it makes the hinge gap wider or narrower.
- **Quarter turn:** one grip of the driver turning a cam or screw a quarter of the way round, then letting go to re-grip.
- **Even gap:** the hinge gap looks the same width at the left hinge as at the right hinge, and the door's hinge edge runs parallel to the board's front edge.
- **Lined up:** the door's left end sits over the left line mark and its right end sits over the right line mark.
- **Flat with no rock:** the door lies down on the table along its whole free edge and does not tip when the right gripper touches it.

- **Fully seated screw:** the screw head sits flat against the plate or the door with no gap under it, and the screw stops advancing.
- **Proud:** the screw head still stands up off the part, with a gap under the head.
- **Upright:** the door stands on its hinge edge, straight up from the table, held by the right gripper.
- **Binds:** the door stops part way through a swing and will not go further without force.
- **Bit empty:** no screw on the screwdriver's bit after the tool lifts away from a driven screw.

Steps

Run Steps 1–9 in order on the one door, then end the episode with Step 10.

Only the tray picks in Step 2.1 and Step 4 depend on the config (Step 3 repeats Step 2.1): the **left gripper** takes each hinge and each plate from the front-left in Config L1, from the back-left in Config L2, and from the front-center in Config M. Every other line is the same in all three configs.

Step 1: Pick up the electric screwdriver

Goal: the electric screwdriver is held upright in the right gripper, bit empty and trigger untouched.

- With the **right gripper**, take the screwdriver from its cradle without touching the trigger.
- With the **left gripper**, steady the screwdriver in the right gripper until the bit points straight down, then open the **left gripper** and move it clear.
- Keep the screwdriver in the **right gripper** through Step 6. Do not re-cradle it between screws.

Check: the bit points straight down, the screwdriver looks upright, and the bit is empty. If it sits tilted or loose, square it again with the left gripper before going on.

Expected state: the right gripper holds the screwdriver upright, the door is untouched on the door spot, and the tray and holder are full.

Step 2: Fit and screw hinge 1 to the door

Goal: hinge 1 sits in cup hole 1 with its pin in the pin hole, held by one fully seated screw.

2.1 Seat the cup

- **IF the parts tray is at the front-left (Config L1):** with the **left gripper**, pick one hinge from the parts tray by the sides of its cup, lift it straight up, and carry it low over the table back and to the right to **cup hole 1**.
- **IF the parts tray is at the back-left (Config L2):** with the **left gripper**, pick one hinge from the parts tray by the sides of its cup, lift it straight up, and carry it low over the table to the

right, clear of the frame board's left end, to **cup hole 1**.

- **IF the parts tray is at the front-center (Config M):** with the **left gripper**, pick one hinge from the parts tray by the sides of its cup, lift it straight up, and carry it low over the table straight back to **cup hole 1**.

Then, in all three:

- With the **left gripper**, lower the cup into **cup hole 1** and drop the **locating pin** into the pin hole beside it.
- Set the hinge so the **arm** points back toward the frame board and lies flat on the door's back face.
- Open the **left gripper** and move it clear once the cup is down in the hole and the pin is in.

Check: the cup is down in the hole, the pin is in the pin hole, the arm points back toward the board and lies flat, and the screw hole in the cup lines up with the pilot hole. If the cup stands proud of the hole or the arm points any other way, lift the hinge with the left gripper and set it again.

2.2 Load one screw onto the bit

- With the **left gripper**, take one screw from the screw holder by the sides of its head, threaded end down.
- With the **left gripper**, raise the screw head to the bit until the bit holds it.
- Open the **left gripper** only once the bit is holding the screw.

Check: the screw hangs from the bit, straight, head in the bit and threaded end down, with the left gripper open and clear. If it hangs crooked or falls, take it again with the left gripper and re-seat it.

2.3 Drive the screw

- With the **left gripper**, press down on the door's back face beside cup hole 1 and hold the door against the table.
- With the **right gripper**, bring the screw tip into the pilot hole in the cup, keeping the screwdriver upright.
- Press the trigger at low speed and drive the screw until the head sits flat on the cup and the screw stops advancing.
- Lift the screwdriver straight up off the driven screw.
- Press the trigger only with the tip in a hole. Do not press it in mid-air and do not lean the screwdriver over to reach.

Expected state: hinge 1 is fixed in cup hole 1, its arm points back toward the board, the bit is empty, and the door has not moved on the door spot.

Check: the screw head is flat on the cup with no gap under it, the screw went in straight, and the hinge does not turn when the left gripper lets go. If the screw is proud or crooked, back it out in reverse with the right gripper, re-seat the cup as in 2.1, and drive it again.

Step 3: Fit and screw hinge 2 to the door

Goal: hinge 2 sits in cup hole 2 the same way, held by one fully seated screw.

- Repeat Steps 2.1, 2.2, and 2.3 on **cup hole 2**, keeping the screwdriver in the **right gripper**.
- Finish hinge 1 completely before starting hinge 2. Do not place both hinges and then drive both screws.

Expected state: both hinges are fixed to the door, both arms point back toward the board and lie flat, two screws are left in the screw holder, and the bit is empty.

Check: both cups are down in their holes, both pins are in their pin holes, both screw heads are flat, and neither hinge turns when touched by the left gripper.

Step 4: Clip a plate onto each hinge arm

Goal: one plate is clicked home under the far end of each arm, with its screw hole facing up.

- With the **left gripper**, pick one plate from the parts tray. **IF Config L1 or M:** carry it low over the table back to the door. **IF Config L2:** carry it low over the table to the right, clear of the frame board's left end, to the door.
- With the **left gripper**, set the plate under the far end of hinge 1's **arm**, screw hole facing up, and press down until it **clicks**.
- With the **left gripper**, do the same on hinge 2's arm with the second plate.
- Press each plate on straight down. Do not twist it onto the arm and do not lever it on with the gripper tip.
- Leave the door on the door spot for this whole step. Do not slide it yet.

Expected state: both plates are clicked home, both sit flat under their arms with their screw holes facing up, and the parts tray is empty.

Check: each plate clicked, sits flat under its arm, and does not rock or slide off when the left gripper nudges it. If a plate rocks or comes away, press it on again with the left gripper until it clicks.

Step 5: Offer the door to the frame board

Goal: the door is slid back until both plates sit flat on their stations with their screw holes over the pilot holes.

- With the **left gripper**, take the door's **free edge** near its left end.
- Slide the door **straight back** along the table toward the frame board in one steady move.

- Keep the door flat on the table and keep it square to the front edge. Do not lift it, do not carry it in the air, and do not swing it round.
- Stop sliding once **plate 1** sits on station 1 and **plate 2** sits on station 2, with each plate's screw hole over its pilot hole.
- Keep the **left gripper** on the door until Step 6 needs it for a screw.

Expected state: the door lies flat in front of the board, both plates rest flat on their stations, both screw holes sit over their pilot holes, and the hinge gap runs the length of the door.

Check: both plates are flat on the board with their holes over the pilot holes, and the hinge gap looks even. If a plate is short of or past its station, slide the door a little with the left gripper until the hole sits over the pilot hole. If a plate came off its arm, slide the door forward again, clip the plate on as in Step 4, then offer the door again.

Step 6: Fix the two plates to the board

Goal: each plate is held down by one fully seated screw, with the door still where Step 5 left it.

6.1 Screw plate 1

- With the **left gripper**, load one screw onto the bit as in Step 2.2, then move clear.
- With the **left gripper**, press down on hinge 1's **arm** just in front of plate 1, holding the plate flat and the door where it is.
- With the **right gripper**, bring the screw tip into plate 1's screw hole, keeping the screwdriver upright.
- Press the trigger at low speed and drive the screw until the head sits flat on the plate and the screw stops advancing.
- Lift the screwdriver straight up off the driven screw.

6.2 Screw plate 2

- Repeat 6.1 on **plate 2**, with the **left gripper** pressing hinge 2's arm just in front of the plate.
- Finish plate 1 before loading the screw for plate 2.

Expected state: both plates are screwed to the board, the door lies flat with an even hinge gap, the screw holder is empty, and the bit is empty.

Check: both screw heads are flat on their plates with no gap under them, both plates sit flat on the board, and the door has not shifted. If a plate lifted or turned as its screw went in, back the screw out in reverse with the right gripper, hold the arm down again with the left gripper, and drive it again. If the door was nudged out of place, back both screws out, offer the door again as in Step 5, then drive both screws.

Step 7: Swap to the hand driver

Goal: the electric screwdriver is back in its cradle and the hand driver is held upright in the right gripper.

- With the **right gripper**, put the electric screwdriver back in its cradle and let go.
- With the **right gripper**, take the **hand driver** from its holder.
- With the **left gripper**, steady the hand driver in the right gripper until its tip points straight down, then open the **left gripper** and move it clear.
- Do not use the electric screwdriver on a cam.

Check: the electric screwdriver is sitting in its cradle, and the hand driver's tip points straight down.

Step 8: Adjust the cams

Goal: the door is lined up with the marks and the hinge gap is even, with both hinges adjusted by the same amount.

- Turn the cams in **quarter turns** with the **right gripper**: set the tip in the cam, turn a quarter, lift the tip clear, set it back, and turn again.
- Turn both hinges' cams by the **same number of quarter turns**, so the door stays square to the board.
- Keep the **left gripper** clear of the door while you look, so the door sits where the hinges hold it.
- Do not force a cam past the point where it stops turning.

8.1 Line the door up

- With the **right gripper**, turn hinge 1's **line cam** one quarter turn, then hinge 2's **line cam** one quarter turn.
- Look at both ends of the door against the **line marks**.
- Repeat until the door's left end sits over the left mark and its right end sits over the right mark.
- Turn the line cams clockwise to move the door right and counter-clockwise to move it left (**unvalidated** — confirm the direction at the station).

Check: both ends of the door sit over their line marks. If the door is skewed, with one end over its mark and the other not, the two hinges have had different numbers of quarter turns: even them up before going on.

8.2 Even the hinge gap

- With the **right gripper**, turn hinge 1's **gap cam** one quarter turn, then hinge 2's **gap cam** one quarter turn.

- Look along the hinge gap from the left hinge to the right hinge.
- Repeat until the gap looks the same width at both hinges and the door's hinge edge runs parallel to the board's front edge.
- Turn the gap cams clockwise to narrow the gap and counter-clockwise to widen it (**unvalidated** — confirm the direction at the station).

Expected state: the door is lined up with both marks, the hinge gap is even, and the door lies flat with no rock.

Check: the gap is even, both ends are still on their marks, and the door lies flat with no rock. If evening the gap pulled the door off its marks, go back to 8.1, line it up again, then check the gap again.

Step 9: Swing-test the door twice

Goal: the door swings up to upright and back flat twice, with both hinges holding and both plates staying down.

- With the **right gripper**, put the hand driver back in its holder and let go.
- With the **left gripper**, press down on the **left end** of the frame board and hold it against the table.
- With the **right gripper**, take the middle of the door's **free edge**.
- Lift the free edge up and back in one steady move until the door stands **upright**, hold it there for 2 seconds, then lower it back flat on the table. That is one swing.
- Do the swing **twice**.
- Lift from the middle of the free edge only. Do not lift from a corner, do not pull the door sideways, and do not push it past upright.
- While the door swings, watch both hinges. Both arms must move together, both plates must stay flat on the board, and no screw head may lift off its part.
- If both swings run clean and the door comes back to the same even gap, go to Step 10.
- If the door **binds**, comes back with an uneven gap, or lifts a plate or a screw head, lower the door flat with the right gripper, take the needed tool back with the right gripper, redo the cams as in Step 8 or re-drive the loose screw as in Step 6, put the tool back, and then run Step 9 again from the start.
- If the same fault comes back after two corrections, stop, end the episode at Step 10 as the door is, and log the rig for a station check.

Expected state: the door lies flat and closed, the hinge gap is even, both ends are on their marks, both plates are screwed flat to the board, and both tools are back in their holders.

Check: the door made two full swings to upright and back, nothing lifted, and the door is closed flat with an even gap.

Step 10: End the episode

Goal: recording ends with the door hung, adjusted, and swing-tested.

1. Confirm the door is closed flat with an even hinge gap, both ends on their line marks, both hinges screwed to the door, both plates screwed to their stations, the parts tray and screw holder empty, both tools back in their holders, and the frame board still square at the back of the table.
2. Return both arms home with grippers open. Homing is the last thing the arms do.
3. Stop recording.

After the episode: reset the workspace

All reset work happens with recording off.

1. With recording off, set the electric screwdriver to reverse and back both plate screws out of the frame board.
2. Unclip both plates from the arms and put them in the parts tray.
3. Back both cup screws out of the door and lift both hinges out of their cup holes.
4. Turn both cams on each hinge back to the middle of their travel, so the next episode starts from the same place. Put both hinges in the parts tray, cups up and arms flat.
5. Set the frame board square at the back of the table, flat and just right of the center. Set the door back on the door spot, back face up, hinge edge toward the board. Set the parts tray for the next episode's config — front-left beside the screw holder (Config L1), back-left (Config L2), or front-center in front of the door (Config M).
6. Stage four screws head-up in the screw holder. Leave all pilot holes empty.
7. Set the electric screwdriver back to forward, low speed, and the validated clutch setting, and put both tools in their holders.
8. Brush dust and swarf out of the cup holes, the pin holes, and every pilot hole.
9. Inspect the parts. Replace the door if a pilot hole is stripped or a cup hole is chewed out. Replace the frame board if a station pilot hole no longer holds a screw. Replace a hinge if its arm is bent, a cam will not hold, or the clip no longer clicks. Replace any screw with a chewed head.
10. Run both Setup checklists before the next episode.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain multiple violations; tag each separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Note on the start position: the violations below were written for Config L1 (parts tray at the front-left, beside the screw holder). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the parts tray is not in the start zone for the config; a gripper reaches across the table for a hinge or a plate; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the left gripper takes every hinge and plate from the tray in its start zone, no arm reaches across the table, and the IF line followed is the one for the config on the table).
- **Coaching note:** look where the tray is before the first reach, then follow that config's IF lines through Steps 2, 3, and 4.

Violation: Tool not squared in the right gripper

- **Visible cue:** the electric screwdriver or the hand driver sits tilted or loose in the right gripper, the left gripper never squares it, and the tool is leaning when it reaches a screw or a cam.
- **SOP rule broken:** Steps 1 and 7 (the left gripper steadies the tool in the right gripper until its tip points straight down, before any screw or cam is touched).
- **Coaching note:** square the tool with the left gripper first. A leaning tool chews the head.

Violation: Tool put down mid-work, or trigger pressed off a screw

- **Visible cue:** the electric screwdriver is re-cradled between screws, is passed to the left gripper, or the trigger runs with the bit in mid-air or against the door.
- **SOP rule broken:** Steps 1–6 (the screwdriver stays in the right gripper through Step 6, and the trigger is pressed only with the tip in a hole).

- **Coaching note:** one pickup, four screws. Trigger only when the tip is in the hole.

Violation: Electric screwdriver used on a cam

- **Visible cue:** a cam is turned with the electric screwdriver, or the hand driver is never taken from its holder before the cams are adjusted.
- **SOP rule broken:** Steps 7 and 8 (cradle the electric screwdriver, take the hand driver, and turn every cam with the hand driver in quarter turns).
- **Coaching note:** cams are turned by hand. Swap the tool before you touch them.

Violation: Hinge cup not seated in its hole

- **Visible cue:** the cup stands proud of the cup hole, the locating pin is beside the pin hole instead of in it, or the hinge turns freely in the hole, and the screw goes in anyway.
- **SOP rule broken:** Step 2.1 (the cup is down in the hole and the pin is in the pin hole before the screw is driven).
- **Coaching note:** cup down, pin in, then look before you drive.

Violation: Hinge arm pointing the wrong way

- **Visible cue:** a hinge is screwed down with its arm pointing toward the front edge, out to a side, or standing up off the door instead of back toward the frame board and flat on the back face.
- **SOP rule broken:** Step 2.1 (set the hinge so the arm points back toward the frame board and lies flat on the door's back face).
- **Coaching note:** arms always point at the board. Check both before the first screw.

Violation: Screw not seated on the bit

- **Visible cue:** the screw hangs crooked from the bit, is held in the left gripper while being driven, or falls off on the way to the hole.
- **SOP rule broken:** Steps 2.2 and 6.1 (the screw hangs straight from the bit, head in the bit and threaded end down, and the left gripper opens only once the bit holds it).
- **Coaching note:** let the bit take the screw, then let go and look at it hanging.

Violation: Door not held while a cup screw is driven

- **Visible cue:** the left gripper is off the door while the right gripper drives a cup screw, and the door slides, lifts, or turns on the door spot.
- **SOP rule broken:** Steps 2.3 and 3 (the left gripper presses the door against the table beside the cup hole while the screw is driven).
- **Coaching note:** left gripper on the door before the trigger, every screw.

Violation: Screw left proud or driven crooked

- **Visible cue:** a screw head stands off the cup or the plate with a gap under it, or the screw goes in leaning, and the step moves on.
- **SOP rule broken:** Steps 2.3, 3, and 6 (drive each screw upright until the head sits flat and the screw stops advancing).
- **Coaching note:** upright tip, drive until it stops, then look under the head.

Violation: Plate not clicked home, or fitted the wrong way

- **Visible cue:** a plate is set on an arm without a click, or it rocks, lifts, or slides off the arm when the door is nudged or slid; or it is set on top of the arm instead of under it, twisted or levered on with the gripper tip, or left with its screw hole facing sideways or down.
- **SOP rule broken:** Step 4 (set the plate under the far end of the arm, screw hole facing up, press straight down until it clicks, and check it does not rock, lift, or slide off).
- **Coaching note:** under the arm, hole up, straight down until it clicks, then nudge it to prove it.

Violation: Door carried instead of slid

- **Visible cue:** the left gripper lifts the door off the table, carries it through the air, or swings it round to bring it to the board.
- **SOP rule broken:** Step 5 (slide the door straight back along the table, flat and square, without lifting it).
- **Coaching note:** keep it flat on the table and push it straight back.

Violation: Door offered out of place, or nudged and screwed anyway

- **Visible cue:** a plate sits short of, past, or off its station when its screw is driven, a screw goes into the board beside the pilot hole, or the door is knocked out of place between Steps 5 and 6 and the driving continues.
- **SOP rule broken:** Steps 5 and 6 (each plate sits flat on its station with its screw hole over the pilot hole, and the door stays where Step 5 left it until both screws are seated).
- **Coaching note:** hole over hole before the trigger. If the door moves, offer it again.

Violation: Plate not held while its screw is driven

- **Visible cue:** the left gripper is off the hinge arm while the right gripper drives a plate screw, and the plate lifts, turns with the screw, or walks off its station.
- **SOP rule broken:** Step 6 (the left gripper presses the arm just in front of the plate, holding the plate flat and the door in place, while the screw is driven).
- **Coaching note:** thumb the arm down first. A free plate always turns with the screw.

Violation: Cam adjustment skipped

- **Visible cue:** the episode goes from the plate screws straight to the swing test or the ending, with no cam ever turned.
- **SOP rule broken:** Steps 8.1 and 8.2 (line the door up with the marks, then even the hinge gap, before the swing test).
- **Coaching note:** every door gets both cams at both hinges, every episode.

Violation: Only one hinge adjusted

- **Visible cue:** cams are turned at one hinge and not the other, or one hinge gets more quarter turns than the other, and the door ends up skewed with one end off its line mark.
- **SOP rule broken:** Step 8 (turn both hinges' cams by the same number of quarter turns).
- **Coaching note:** a quarter at one hinge, a quarter at the other, then look. Never chase one hinge.

Violation: Cam forced or over-turned

- **Visible cue:** the right gripper turns a cam more than a quarter turn in one grip, keeps turning after the cam has stopped, or the driver tip slips out and scars the arm.
- **SOP rule broken:** Step 8 (turn cams in quarter turns, letting go to re-grip, and never force a cam past where it stops).
- **Coaching note:** small bites, let go, re-grip. Stop when it stops.

Violation: Cams adjusted in the wrong order

- **Visible cue:** the gap cams are turned before the door has been lined up with the marks, or the door is left off its marks after the gap is evened and nothing goes back to line it up.
- **SOP rule broken:** Steps 8.1 and 8.2 (line the door up first, then even the gap, and go back to 8.1 if evening the gap pulls the door off its marks).
- **Coaching note:** line first, gap second, then look at both again.

Violation: Door held while its position is judged

- **Visible cue:** the left gripper is holding, pressing, or propping the door while the line marks or the hinge gap are being judged, so the hinges are not what is holding the door.
- **SOP rule broken:** Step 8 (keep the left gripper clear of the door while you look, so the door sits where the hinges hold it).
- **Coaching note:** let go, then look. A held door tells you nothing.

Violation: Swing test skipped or cut short

- **Visible cue:** the episode ends with fewer than two swings, or the door is lifted only part way instead of up to upright.
- **SOP rule broken:** Step 9 (lift the door to upright, hold for 2 seconds, lower it flat, and do this twice).

- **Coaching note:** two swings, all the way to upright, every episode.

Violation: Door swung the wrong way

- **Visible cue:** the right gripper lifts the door by a corner or a side edge, drags it sideways, jerks it up, or pushes it past upright.
- **SOP rule broken:** Step 9 (take the middle of the free edge and lift up and back in one steady move, stopping at upright).
- **Coaching note:** middle of the free edge, one steady lift, stop at upright.

Violation: Frame board not held during the swing test

- **Visible cue:** the left gripper is off the frame board while the door swings, and the board lifts, slides, or turns on the table.
- **SOP rule broken:** Step 9 (the left gripper presses the left end of the frame board against the table through both swings).
- **Coaching note:** pin the board first. The swing pulls on it every time.

Violation: Swing fault ignored, or correction not re-tested

- **Visible cue:** the door binds, comes back with an uneven gap, or lifts a plate or a screw head, and the episode ends anyway; or a cam or screw is corrected after a fault and the episode ends with no fresh pair of swings.
- **SOP rule broken:** Step 9 (on a bind, an uneven gap, or a lifting plate or screw, correct it and run Step 9 again from the start).
- **Coaching note:** every fix earns two more swings.

Violation: Work skipped or done out of order

- **Visible cue:** the steps run out of sequence — both hinges placed before either screw is driven, the door offered before both plates are clipped on, a cam turned before both plate screws are seated, the swing test run before the cams — or the episode ends with only one plate screwed down.
- **SOP rule broken:** Steps 2–9 (hinge 1, hinge 2, clip both plates, offer, screw both plates, cams, swing test, in that order, each finished before the next starts).
- **Coaching note:** the order is the task. Say the next step's name before you move.

Violation: Wrong arm used

- **Visible cue:** an action assigned to one gripper is done by the other, including the left gripper holding a tool, driving a screw, turning a cam, or swinging the door, or the right gripper fetching parts from the tray, sliding the door back, or holding the board down.
- **SOP rule broken:** Steps 1–9 (the right gripper holds both tools, drives all four screws, turns all four cams, and swings the door; the left gripper presents the parts, holds the door and the arms down, slides the door back, and holds the board).

- **Coaching note:** right gripper does the tools and the swing, left gripper feeds and holds.

Violation: Part dropped or knocked off its spot

- **Visible cue:** a hinge, plate, or screw is dropped on the board, the door, the table, or the floor; the parts tray or screw holder is tipped or pushed off its spot; or the frame board is knocked out of square.
- **SOP rule broken:** Steps 2–9 (keep the parts, the tray, the holder, and the frame board on their spots through the hang).
- **Coaching note:** work slower and lower over the table, and keep the carry short.

Violation: Wrong episode ending

- **Visible cue:** recording stops before the end state is confirmed, a tool is left out of its holder, an arm is not home, a gripper is closed, or an arm does something else after homing.
- **SOP rule broken:** Step 10 (confirm the end state, return both arms home with grippers open as their final action, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

Failures not caused by how the task was run are system issues. Log and discard the episode rather than tagging them as SOP violations.

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.
- An arm or gripper fails, drifts, or reports a motor error.
- The electric screwdriver's clutch slips or its battery dies mid-screw.
- A pilot hole in the door or the board is stripped, so a correctly driven screw will not pull down flat.
- A cup hole is chewed out, so a correctly seated cup still turns.
- A hinge arm is bent, or a cam turns without moving the door, so a correctly adjusted hinge cannot hold the gap.
- A plate's clip is worn, so a plate pressed until it clicks still comes off the arm.
- A screw head is chewed from the factory, so the bit cannot hold or drive it.

Annotation subtasks (from SOP)

1. Pick up the electric screwdriver and square it in the right gripper
2. Seat hinge 1 in cup hole 1 and drive its screw
3. Seat hinge 2 in cup hole 2 and drive its screw
4. Clip a plate onto each hinge arm

5. Slide the door back to offer it to the frame board
6. Drive the screw into plate 1, then into plate 2
7. Swap the electric screwdriver for the hand driver
8. Turn the line cams until the door lines up with the marks
9. Turn the gap cams until the hinge gap is even
10. Swing the door up to upright and back flat, twice
11. Confirm the end state, return both arms home, and end the episode